

SQUAMOUS CELL CARCINOMA IN SITU (BOWEN DISEASE)

AT A GLANCE

- Bowen disease (BD) is squamous cell carcinoma (SCC) in situ, with the potential to progress to invasive SCC.
- Etiologic factors include ultraviolet radiation exposure, chronic arsenicism, previous psoralen plus ultraviolet A (PUVA) therapy, immunosuppression, ionizing radiation, and human papillomavirus (HPV) infection.
- HPV DNA has been detected in up to 30% of extragenital BD lesions, with HPV-16 commonly found in anogenital, finger, and periungual cases.
- BD is more common in sun-exposed areas (e.g., head, neck, lower legs) and rare in individuals with darker pigmentation, supporting a role for UV radiation.
- Increased incidence is observed in organ transplant recipients due to long-term immunosuppression.
- Clinical variants include pigmented, intertriginous, periungual, and subungual BD.
- Histopathologically, BD shows full-thickness epidermal atypia with involvement of adnexal structures; the basement membrane remains intact.
- Lesions may progress to invasive carcinoma in 3% to 5% of cases.
- Treatment options include excision, Mohs micrographic surgery (allowing complete margin assessment), and topical therapies for difficult-to-treat areas.
- Curettage may miss adnexal involvement and is less reliable for definitive clearance.

Clinical Findings

BD typically presents as a discrete, slowly enlarging, pink to erythematous thin plaque with well-demarcated, irregular borders and overlying scale or crust, resembling a psoriatic plaque. The surface may be hyperkeratotic or verrucous. A pigmented variant occurs in less than 2% of cases.

Lesions can measure up to several centimeters in diameter, and multiple lesions are common. Predilection sites include sun-exposed areas such as the head, neck, and lower extremities, though any skin site can be affected.

Clinical Variants

- **Intertriginous BD:** May present as an oozing, erythematous, dermatitis-like plaque or as a pigmented patch or plaque, often in axillary, inguinal, or genital regions.
- **Periungual and subungual BD:** Appears as an erythematous, scaly plaque around the nail fold, crusted erosion, nail discoloration, onycholysis, verrucous growth, or nail plate destruction.
- **Mucosal BD:** Presents as verrucous or polypoid papules and plaques, erythroplakia, or a velvety erythematous plaque. Erythroplasia of Queyrat, a form of genital mucosal BD, is discussed separately.

Histopathology

The hallmark of BD is full-thickness epidermal atypia involving keratinocytes from the stratum corneum to the basal layer, including intraepidermal adnexal structures. The basement membrane remains intact, distinguishing it from invasive SCC.

Key histologic features include: - Parakeratosis and hyperkeratosis - Acanthosis with loss of normal epidermal maturation - Marked cellular pleomorphism and increased mitotic activity - Disorganized epidermal architecture

Adnexal involvement is common and must be evaluated carefully during treatment to ensure complete eradication.